02/04/2025
Morning

Aakash
Medical | IIT-JEE | Foundations

Corporate Office : AESL, 3rd Floor, Incuspaze Campus-2, Plot-13, Sector-18, Udyog Vihar, Gurugram, Haryana-122015

Join our Youtube channel for JEE Main Memory Based Paper Live Discussion

Memory Based
Answers & Solutions
for
JEE (Main)-2025 (Online) Phase-2

(Physics, Chemistry and Mathematics)

Time : 3 hrs. **M.M. : 300**

IMPORTANT INSTRUCTIONS:

(1) The test is of **3 hours** duration.

(2) This test paper consists of 75 questions. Each subject (PCM) has 25 questions. The maximum marks are 300.

(3) This question paper contains **Three Parts**. **Part-A** is Physics, **Part-B** is Chemistry and **Part-C** is **Mathematics**. Each part has only two sections: **Section-A** and **Section-B**.

(4) **Section - A :** Attempt all questions.

(5) **Section - B :** Attempt all questions.

(6) **Section - A (01 – 20)** contains 20 multiple choice questions which have **only one correct answer**. Each question carries **+4 marks** for correct answer and **–1 mark** for wrong answer.

(7) **Section - B (21 – 25) contains** 5 **Numerical value** based questions. The answer to each question should be rounded off to the **nearest integer**. Each question carries **+4 marks** for correct answer and **–1 mark** for wrong answer.



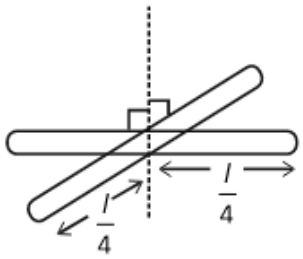
PHYSICS

SECTION - A

Multiple Choice Questions: This section contains 20 multiple choice questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

Choose the correct answer:

1. The moment of inertia of a uniform rod of mass m and length l is α when rotated about an axis passing through centre and perpendicular to the length. If the rod is broken into equal halves and arranged as shown, then the moment of inertia about the given axis is



(1) 2α

(2) $\frac{\alpha}{2}$

(3) 4α

(4) $\frac{\alpha}{4}$

Answer (4)

Sol. $I_i = \frac{ml^2}{12} = \alpha$

$$I_f = 2\frac{\left(\frac{m}{2}\left(\frac{l}{2}\right)^2\right)}{12} = \frac{ml^2}{48}$$

$$I_f = \frac{\alpha}{4}$$

2. Two point charges q and $9q$ are placed at distance of l from each other. Then the electric field is zero at a

(1) Distance $\frac{l}{4}$ from charge $9q$

(2) Distance $\frac{3l}{4}$ from charge q

(3) Distance $\frac{l}{3}$ from charge $9q$

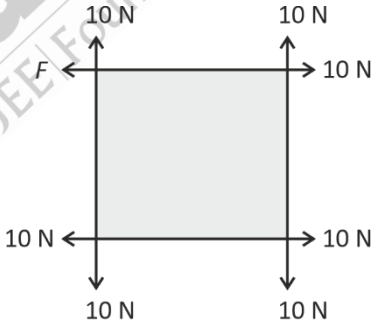
(4) Distance $\frac{l}{4}$ from charge q

Answer (4)

Sol. $\frac{Kq}{x^2} = \frac{K9q}{(l-x)^2} \Rightarrow 3x = l - x$

$$x = \frac{l}{4}$$

3. A square shape lamina of mass M kg is at rest. Find value of F (in N).



(1) 10 N (2) 15 N

(3) 20 N (4) 30 N

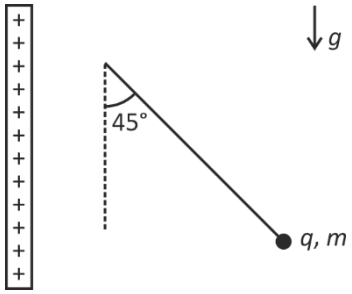
Answer (1)

Sol. In equilibrium net force most zero.

So, F = 10 N



4. The figure shows an infinite plane having uniform charge density σ and a small charged particle having charge *q* and mass *m* suspended by a light insulating thread. Find σ if the charge is in equilibrium.



(1) $\frac{2\varepsilon_0 mg}{q}$ (2) $\frac{\varepsilon_0 mg}{2q}$

(3) $\frac{2q}{\varepsilon_0 mg}$ (4) $\frac{2q\varepsilon_0}{mg}$

Answer (1)

Sol. $\tan\theta\frac{q\left(\frac{\sigma}{2\varepsilon_0}\right)}{mg}=\tan45°$

$\Rightarrow \sigma=\frac{2\varepsilon_0 mg}{q}$

5. Find the ratio of $\left(\gamma=\frac{C_p}{C_v}\right)$ for two gases having degree of freedoms $f = 3$ & $f = 5$.

(1) $\frac{21}{25}$

(2) $\frac{3}{7}$

(3) $\frac{25}{21}$

(4) $\frac{7}{3}$

Answer (3)

Sol. $\gamma=1+\frac{2}{f}$

$\gamma_1=1+\frac{2}{3}=\frac{5}{3}$

$\gamma_2=1+\frac{2}{5}=\frac{7}{5}$

$\frac{\gamma_1}{\gamma_2}=\frac{25}{21}$

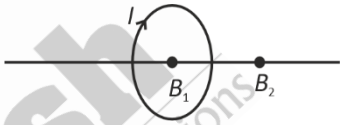
6. A current carrying wire is in form of a circle of radius *R*. Find ratio of magnetic field at centre to the magnetic field at axial point at a distance R from it's centre.

(1) 2 (2) $2\sqrt{2}$

(3) $\sqrt{3}$ (4) $\frac{1}{\sqrt{2}}$

Answer (2)

Sol.



$B_1=\frac{\mu_0 I}{2R}$

$B_2=\frac{\mu_0 IR^2}{2\cdot(2R^2)^{3/2}}=\frac{\mu_0 I}{2\cdot2\sqrt{2}R}$

$\frac{B_1}{B_2}=2\sqrt{2}$

7. Two SHMs are superimposed on a particle moving along *x* axis as $x_1 = A \sin \omega t$ and $x_2 = A \sin(\omega t + \pi/3)$ then acceleration of particle is (given $\omega = 5$)

(1) $-25\ A \sin(5t + \pi/3)$

(2) $-25\ A \sin(5t + \pi/6)$

(3) $-25\sqrt{3}\ A \sin(5t + \pi/6)$

(4) $25\sqrt{3}\ A \sin(5t - \pi/6)$

Answer (3)



Sol. $x = x_1 + x_2 = 2A\sin(\omega t + \pi/6)$

$$\cos\left(\frac{\pi}{6}\right) = \sqrt{3}\,A\sin(\omega t + \pi/6)$$

$$\Rightarrow \quad a = -\sqrt{3}\,A\omega^2\sin(\omega t + \pi/6)$$

$$= -25\sqrt{3}\,A\sin(5t + \pi/6)$$

8. What is the ratio of radius of nth orbit in H, He^+ and Li^{+2}? (assume Bohr model is applicable)

(1) 6 : 3 : 1

(2) 6 : 3 : 2

(3) 3 : 6 : 2

(4) 4 : 3 : 2

Answer (2)

Sol. $r_n = \dfrac{n^2 r_0}{Z}$

$$r_H : r_{He^+} : r_{Li^{+2}} = 1 : \frac{1}{2} : \frac{1}{3}$$

$$= 6 : 3 : 2$$

9. Find the dimension of $\dfrac{a}{b^2}$ from the given formula $\left(P + \dfrac{a}{v^2}\right)(v - b) = RT$ where symbols have their usual meaning.

(1) ML^3T^{-2}

(2) $ML^{-1}T^{-2}$

(3) ML^3L^{-1}

(4) MLT

Answer (2)

Sol. $a \equiv Pv^2$

$b \equiv v$

$$\frac{a}{b^2} = \frac{Pv^2}{v^2} \equiv P \equiv \frac{MLT^{-2}}{L^2}$$

10. Match the List-I with the List-II.

	List-I		List-II
(i)	Coefficient of viscosity	(a)	$M^0L^0T^0$
(ii)	Strain	(b)	$M^{-1}LT^2$
(iii)	Compressibility	(c)	$ML^{-2}T^{-2}$
(iv)	Pressure gradient	(d)	$ML^{-1}T^{-1}$

(1) (i)-(a), (ii)-(c), (iii) -(d), (iv)-(b)

(2) (i)-(d), (ii)-(a), (iii) -(b), (iv)-(c)

(3) (i)-(b), (ii)-(d), (iii) -(c), (iv)-(a)

(4) (i)-(c), (ii)-(b), (iii) -(d), (iv)-(a)

Answer (2)

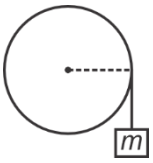
Sol. [Coefficient of viscosity] = $MT^{-1}T^{-1}$

$$[\text{Compressibility}] = \frac{1}{[\text{Bulk Modulus}]} = M^{-1}LT^2$$

[Strain] = dimensionless

$$[\text{Pressure gradient}] = \frac{[\text{Pressure}]}{[\text{Length}]} = ML^{-2}T^{-2}$$

11. The figure shows a disc of mass 5 kg and radius 10 cm having axis fixed and free to rotate about its axis. A 2 kg block is suspended with the help of a string wound around the disc and released from rest. The angular velocity of the disc when the block moves by 0.2 m is (Take g = 10 ms^2)



(1) $\dfrac{40}{9}$ rad/s (2) $\dfrac{40}{3}$ rad/s

(3) $\dfrac{30}{7}$ rad/s (4) $\dfrac{28}{5}$ rad/s

Answer (2)



Sol. From work energy theorem

$$mgh = \frac{1}{2}mv^2 + \frac{1}{2}\left(\frac{MR^2}{2}\right)\omega^2$$

Also $v = \omega R$

$$\Rightarrow \quad \omega = \sqrt{\frac{4mgh}{(2m+M)R^2}}$$

$$= \frac{40}{3} \text{ rad/s}$$

12. Speed of a boat in still water is 27 km/hr and speed of river flow is 9km/hr. If boat rows at angle 150° with the river flow and it takes 30 seconds to cross the river. Find with of the river.

(1) 56 m

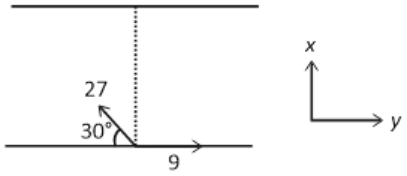
(2) 100 m

(3) 225 m

(4) 112.5 m

Answer (4)

Sol.

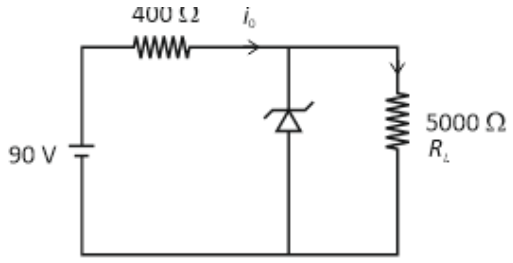


$V_y = 27 \times \frac{1}{2}$ km/h

$$W = \frac{27}{2} \times \frac{5}{18} \times 30$$

W = 112.5 m

13. The figure shows an electrical circuit with a zener diode having a rated voltage V_z = 30 V. Find the current through the diode in mA



(1) 6 mA

(2) 150 mA

(3) 144 mA

(4) 154 mA

Answer (3)

Sol. $V_{(across)}$ Zener = 30 V $\quad i$ through $= \frac{30}{5000} = 6$ mA

V_{across} R = 90 – 30 V = 60 V $\quad i$ through $R = \frac{60}{400} = 150$ mA

150mA = $I_z + {}^{i}R_z$ $\quad I_z$ = 144 mA

150 mA = I_z + 6 mA

14. Correct relation between permeability of vacuum μ_0, relative permeability μ_r, permeability of medium μ and magnetic susceptibility χ is

(1) $\mu = \chi\mu_0$

(2) $\mu = (\mu_{r+1})\,\mu_0$

(3) $\mu = (1 + \chi)\,\mu_0$

(4) $\mu = \chi(\mu_r + 1)$

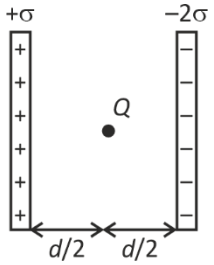
Answer (3)

Sol. $\mu_r = 1 + \chi$ $\quad \mu = \mu_r\,\mu_0$

$\mu = (1 + \chi)\,\mu_0$



15. Two uniformly charged sheets are shown in the figure. Find net force on the charge Q placed symmetrically between the plates.

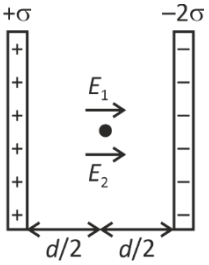


(1) $\frac{3\sigma Q}{\varepsilon_0}$ (2) $\frac{2\sigma Q}{\varepsilon_0}$

(3) $\frac{3}{2}\frac{\sigma Q}{\varepsilon_0}$ (4) Zero

Answer (3)

Sol.



$F = E_1 + E_2$

$= \frac{\sigma}{2\varepsilon_0} + \frac{2\sigma}{2\varepsilon_0}$

$= \frac{3}{2}\frac{\sigma}{\varepsilon_0}$

$F = \frac{3}{2}\frac{\sigma Q}{\varepsilon_0}$

16. In a single slit diffraction using light of wavelength λ, the 2nd minima is formed at θ_1 and 3rd maxima is at θ_2. If $\theta_1 + \theta_2 = 30°$, then the slit width is

(1) $\frac{66\lambda}{\pi}$ (2) $\frac{22\lambda}{\pi}$

(3) $\frac{33\lambda}{\pi}$ (4) $\frac{11\lambda}{\pi}$

Answer (3)

Sol. 2nd minima is formed at $\theta_1 = \frac{2\lambda}{a}$ $\left(\theta = \frac{n\lambda}{a}\right)$

3rd maxima is formed at $\theta_2 = \frac{7\lambda}{2a}$ $\left(\theta = (2m+1)\frac{n\lambda}{a}\right)$

$\theta_1 + \theta_2 = \frac{\pi}{6} \Rightarrow \frac{2\lambda}{a} + \frac{7\lambda}{2a} = \frac{\pi}{6}$ or $a = \frac{33\lambda}{\pi}$

17. A wine having length 2 m having yang modulus $Y = 2 \times 10^{11}$ N/m^2. Axial strain is 0.2 and Poisson ratio is 0.01 for a deformation. Find energy density for elastic energy stored in rod.

(1) 2×10^9 J/m^3 (2) 4×10^9 J/m^3

(3) 10^9 J/m^3 (4) 5×10^8 J/m^3

Answer (2)

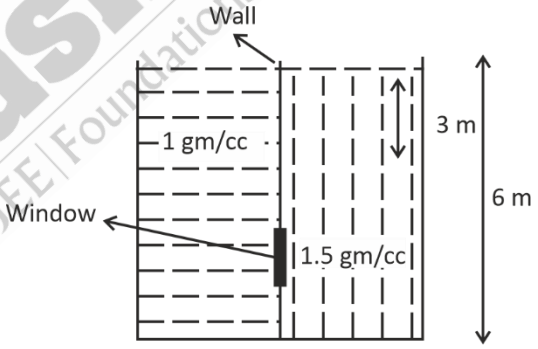
Sol. Energy stored per unit volume will

$= \bar{\varepsilon} = \frac{1}{2}\sigma\varepsilon = \frac{1}{2}\gamma\varepsilon^2$

$= \frac{1}{2} \times 2 \times 10^{11} \times 0.2 \times 0.2$

$= 4 \times 10^9$ J/m^3

18. A concrete wall separates two liquids of densities 1 gm/cc and 1.5 gm/cc as shown. There is a square window of area 100 cm^2 of wood instead of concrete. Assuming no friction, find extend force needed to hold the window.



(1) 400 N (2) 300 N

(3) 150 N (4) 250 N

Answer (3)

Sol. $F = |p_1 - p_2|A = |\rho_1 - \rho_2|ghA$

$= 0.5 \times 10^3 \times 10 \times 3 \times 100 \times 10^{-4}$

$= 150$ N

19.

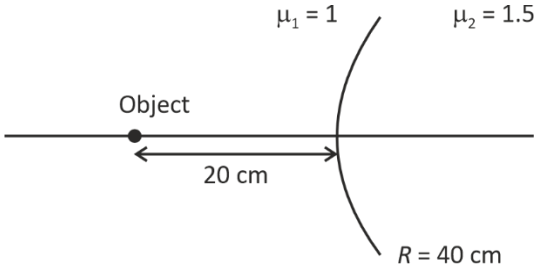
20.



SECTION - B

Numerical Value Type Questions: This section contains 5 Numerical based questions. The answer to each question should be rounded-off to the nearest integer.

21. The figure shows a spherical surface of radius 40 cm separating two mediums of refractive indices $\mu_1 = 1$ and $\mu_2 = 1.5$. A point object in the medium μ_1 at a distance 20 cm from the separation forms an image whose distance from the separation is x cm, where x is



Answer (40)

Sol. $\mu_1 = 1$, $\mu_2 = 1.5$, $R = +40$ cm, $u = -20$ cm

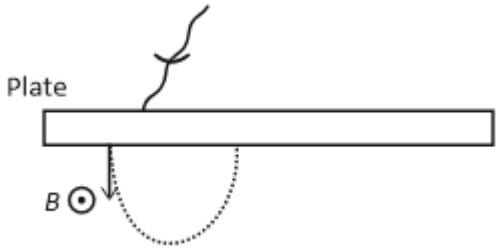
$$\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$$

$$\frac{1.5}{v} - \frac{1}{-20} = \frac{1.5-1}{+40}$$

$$\frac{1.5}{v} = \frac{1}{80} - \frac{1}{20}$$

$$\Rightarrow \quad v = -40 \text{ cm}$$

22. Photons of wavelength λ is incident on a metal of work function ϕ, there exist a magnetic field B, perpendicular to the initial velocity of photoelectron. If radius of photoelectron is $\frac{1}{eB}\sqrt{\alpha m\left(\frac{hc}{\lambda} - \phi\right)}$. Find value of α.



Answer (2)

Sol. $\frac{hc}{\lambda} - \phi = \frac{1}{2}mv^2$

$$R = \frac{mv}{qB} = \frac{m}{eB}\sqrt{\left(\frac{2hc}{\lambda} - 2\phi\right)\frac{1}{m}}$$

$$R = \frac{1}{eB}\sqrt{2m\left(\frac{hc}{\lambda} - \phi\right)}$$

23. Particle is moving with constant speed $V_1 = 5$ m/s for distance x and remaining $\frac{3x}{2}$ distance with constant speed V_2. Find V_2 (m/s) is average speed for the journey is $\frac{5}{7}$ m/s.

Answer (10)

Sol. $V_{av} = \frac{\text{Total distance}}{\text{Total time}}$

$$\text{Total distance} = x + \frac{3x}{2}$$

$$= \frac{5x}{2}$$

$$\text{Total time} = \frac{x}{5} + \frac{3x}{2V_2}$$

$$\frac{50}{7} = \frac{\frac{5}{2}x}{\frac{x}{5} + \frac{3x}{2V_2}}$$

$$50\left(\frac{1}{5} + \frac{3}{2V_2}\right) = \frac{7\times 5}{2}$$

$$20 + \frac{150}{V_2} = 35$$

$$V_2 = 10$$

24.

25.



CHEMISTRY

SECTION - A

Multiple Choice Questions: This section contains 20 multiple choice questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

1. Which of the following is correct order of basic strength of amines in aqueous medium

(1) $CH_3NH_2 > (CH_3)_2NH > (CH_3)_3N > NH_3$

(2) $(CH_3)_2NH > CH_3NH_2 > (CH_3)_3N > NH_3$

(3) $CH_3NH_2 > NH_3 > (CH_3)_2NH > (CH_3)_3N$

(4) $(CH_3)_3N > (CH_3)_2NH > CH_3NH_2 > NH_3$

Answer (2)

Sol. Basic strength of methyl amine follows the order

$2° > 1° > 3° > NH_3$ (In aqueous medium)

2. Which of the following statement(s) is/are correct for the adiabatic process?

A. Molar heat capacity is zero.

B. Molar heat capacity is infinite.

C. Work done on gas is equal to increase in internal energy

D. The increase in temperature results in decrease in internal energy

(1) A & C only (2) B & C only

(3) A & D only (4) C and D only

Answer (1)

Sol. A and C statements are correct $\Delta U = q + w$

For adiabatic process $q = 0$

$\therefore \Delta U = w$

3. In group 17, which property does not follow regular trend?

(1) Electron affinity (2) Ionisation energy

(3) Covalent radii (4) Ionic radii

Answer (1)

Sol. 'F' despite having smaller size, has smaller electron affinity than Cl. It is due to $e^- - e^-$ repulsion in F.

4. Out of the following species, which one is anti-aromatic?

(1) 

(2) 

(3) 

(4) 

Answer (2)

Sol. A species is antiaromatic if it has $(4n)\pi$ electrons delocalised over the entire cyclic ring, where n = 1, 2, 3...



is anti-aromatic as it has 4π electrons delocalised over five membered cyclic ring.

5. In following reaction sequence product C is

$$But-2-ene + Br_2 \xrightarrow{CCl_4} A \xrightarrow[excess]{NaNH_2} B \xrightarrow{Hg^{2+}/H_2SO_4} C$$

Identity C:

(1) $CH_3-\overset{\overset{O}{\|}}{C}-CH_2-CH_3$

(2) $CH_3-CH_2-CH_2-CHO$

(3) $CH_3-CH_2-CH_2-CH_2-OH$

(4) $HC(=O)-CH_2-CH_2-C(=O)-H$

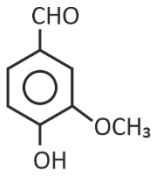
Answer (1)

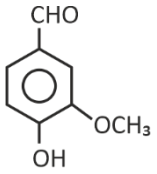


Sol. $CH_3-CH=CH-CH_3 \xrightarrow[CCl_4]{Br_2} CH_3-\underset{Br}{CH}-\underset{Br}{CH}-CH_3$ (A)

$\xrightarrow{NaNH_2(excess)}$ $CH_3-C\equiv C-CH_3$ (B)

$\xrightarrow[H_2SO_4]{Hg^{2+}}$ $CH_3-\overset{O}{\overset{\|}{C}}-CH_2-CH_3$ (C)

6. Statement-I :  reacts with NaOH to give a compound which gives positive Tollens test.

Statement-II :  reacts with NaOH to give self aldol condensation reaction.

In the light of above statements, choose the correct option.

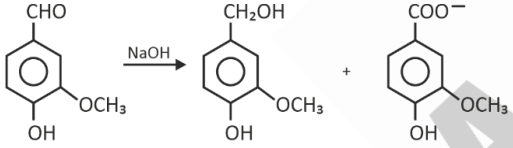
(1) Statement-I and Statement-II both are correct

(2) Statement-I is correct and Statement-II is incorrect

(3) Statement-II is correct and Statement-I is incorrect

(4) Statement-I and Statement-II both are incorrect

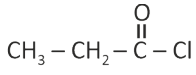
Answer (4)

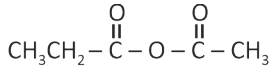
Sol. 

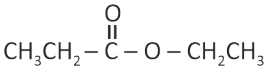
It gives Cannizarro reaction.

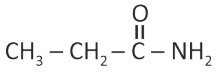
None of the products gives Tollens test

7. Arrange the following compounds in the decreasing order of their rate of hydrolysis in presence of acid as a catalyst.


(a)


(b)


(c)


(d)

(1) a > b > c > d (2) b > a > c > d

(3) c > a > b > d (4) d > c > b > a

Answer (1)

Sol. The rate of acid catalysed hydrolysis of acid derivatives is decided by leaving group ability. Lower the basic strength greater the leaving group ability, higher will be the rate of hydrolysis.

$R-\overset{O}{\overset{\|}{C}}-L + H_2O \underset{}{\overset{H^+}{\rightleftharpoons}} R-\overset{O}{\overset{\|}{C}}-OH + L^- + H^+$

	Acid derivative	Leaving group
(a)	$CH_3-CH_2-\overset{O}{\overset{\|}{C}}-Cl$	Cl^-
(b)	$CH_3CH_2-\overset{O}{\overset{\|}{C}}-O-\overset{O}{\overset{\|}{C}}-CH_3$	$CH_3-\overset{O}{\overset{\|}{C}}-O^-$
(c)	$CH_3CH_2-\overset{O}{\overset{\|}{C}}-O-CH_2CH_3$	$CH_3CH_2O^-$
(d)	$CH_3-CH_2-\overset{O}{\overset{\|}{C}}-NH_2$	NH_2^-

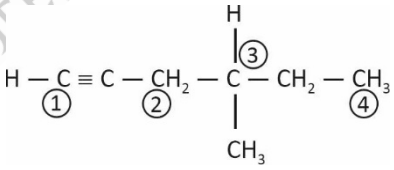
Order of basic strength of leaving groups

$Cl^- < CH_3-\overset{O}{\overset{\|}{C}}-O^- < CH_3CH_2O^- < NH_2^-$

$\therefore$ Decreasing order rate of hydrolysis of the given acid derivatives

a > b > c > d

8. Which of the following carbon atom, forms the least stable and most stable free radical, respectively.



(1) 1, 3 (2) 1, 4

(3) 1, 2 (4) 3, 4

Answer (1)

Sol.

- Least stable free radical formed on carbon marked as (1), because that carbon is sp hybridised.
- Most stable free radical formed on carbon marked as (3) because it has 7 α H atoms available for hyperconujugation

THE LEGACY OF SUCCESS CONTINUES
JEE Main (Session-1) 2025
4 STATE TOPPERS | 70+ 100 PERCENTILERS (PHY. | CHEM. | MATHS) | 1000+ 99 PERCENTILERS & ABOVE | 4000+ 95 PERCENTILERS & ABOVE
100 Percentile | Shreyas Lohiya | PSID: 00003389699
100 Percentile | Harsh Jha | PSID: 00014863322
99.99 Percentile | Devya Rustagi | PSID: 00014768785
99.99 Percentile | Amogh Bansal | PSID: 00014769016
OUR JEE CHAMPIONS
Chirag Falor 4 Year Classroom 1 AIR JEE (Adv.) 2020
Tanishka Kabra 4 Year Classroom ALL INDIA FEMALE TOPPER 1 AIR-16 CRL JEE (Adv.) 2022
Sanvi Jain 4 Year Classroom ALL INDIA FEMALE TOPPER 1 AIR-34 CRL JEE (Main) 2024

9. Given below are two statements.

Statement I : All naturally occurring amino acids are optically active except glycine.

Statement : All amino acids are optically active. In light of the above statements, choose the most appropriate option

(1) Statement I is true, Statement II is false

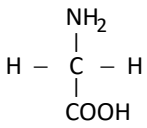
(2) Both statement I and statement II are true

(3) Both statement I and statement II are false

(4) Statement I is false while statement II is true

Answer (1)

Sol. Glycine is the only optically inactive naturally occurring amino acid.



10. Consider the following statements

Statement-1 : Metallic radius of Al is less than that of Ga.

Statement-2 : Ionic radius of Al^{3+} is less than that of Ga^{3+}

(1) Statement-I and Statement-II both correct

(2) Statement-I and Statement-II both incorrect

(3) Statement-I is incorrect Statement-II is correct

(4) Statement-I and Statement-II, both one incorrect

Answer (3)

Sol. Metallic radius of Ga is less than that of Al. This can be understood from the variation in the inner core of the electronic configuration. The presence of additional 10 *d*-electrons offer only poor screening effect for the outer electrons from the increased nuclear charge in gallium.

Ga(135 pm) < Al (143 pm)

Ga^{3+} (62 pm) > Al^{3+} (53.5 pm)

11. Out of the following, XeF_2, NH_3, NF_3, ClF_3, SF_4 and SO_2 Compound having non zero dipole moment and central atom having maximum lone pair of electrons, respectively are :-

(1) ClF_3 and SF_4 (2) XeF_2 and SO_2

(3) ClF_3 and XeF_2 (4) NH_3 and NF_3

Answer (3)

Sol. $XeF_2 \rightarrow$ dipole moment = 0

NH_3, NF_3, ClF_3, SF_4, SO_2 $\rightarrow$ Non-zero dipole moment

Number of lone pairs on central atom

$XeF_2 \Rightarrow 3$

$NH_3 \Rightarrow 1$

$NF_3 \Rightarrow 1$

$ClF_3 \Rightarrow 2$

$SF_4 \Rightarrow 1$

$SO_2 \Rightarrow 1$

$\therefore$ XeF_2 has maximum number of lone pairs

12. In AX_4B, if A is p-block group-18 element, X is most electronegative element and B is second most electronegative element, then shape of compound will be

(1) Square pyramidal

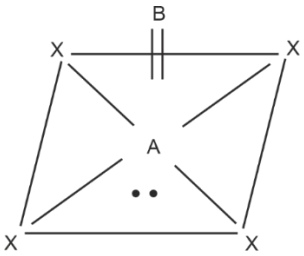
(2) Trigonal bipyramidal

(3) Pentagonal Planar

(4) Distorted pentagonal pyramidal

Answer (1)

Sol.



e.g., $XeOF_4$

Oxygen is second most electronegative element



13. Among Fe, Mn, Co and Cr, element having highest value of $E^{o}_{M^{3+}/M^{2+}}$ form a complex $[M(CN)_6]^{3-}$, then find the number of electrons in e_g set of orbital

(1) 0 (2) 1

(3) 2 (4) 4

Answer (1)

Sol. $E^{o}_{Fe^{3+}/Fe^{2+}} = 0.77$ V

$E^{o}_{Co^{3+}/Co^{2+}} = 1.97$ V

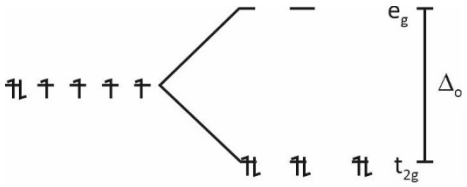
$E^{o}_{Cr^{3+}/Cr^{2+}} = -0.41$ V

$E^{o}_{Mn^{3+}/Mn^{2+}} = 1.57$ V

Cobalt having maximum value of $E^{o}_{Co^{3+}/Co^{2+}}$ i.e. 1.97 V

Complex compound should be $[Co(CN)_6]^{3-}$

$Co^{3+} \Rightarrow [Ar]3d^6 4s^0$



Δ_o > P as CN^- is a SFL with Co^{3+} ion

$t_{2g}^6 e_g^0$

Number of electrons in e_g set of orbitals = 0

14. Statement I : If paring energy (P) is greater than Δ_o then high spin complex is formed and if P < Δ_o, then low spin complex is formed.
Statement II : If Δ_t < P, then high spin complex is formed.
(Δ_o – CFSE in octahedral field, Δ_t – CFSE in tetrahedral field)

(1) Both statements I and II are correct
(2) Statement I is correct Statement II is incorrect
(3) Statement I is incorrect Statement II is correct
(4) Both statements I and II are incorrect

Answer (1)

Sol. If pairing energy is higher, then electrons do not pair- results in high spin complex. In tetrahedral field Δ_t < P.

15.

16.

17.

18.

19.

20.

SECTION - B

Numerical Value Type Questions: This section contains 5 Numerical based questions. The answer to each question should be rounded-off to the nearest integer.

21. Vapour pressure of pure liquid A is 200 mm Hg. If 1 mol of A and 3 mol of B are mixed. Assuming solution to be ideal, find the vapour pressure of pure liquid 'B', if total vapour pressure of solution is 500 mm Hg

Answer (600)

Sol. For an ideal solution $P_S = P_A^{\circ} X_A + P_B^{\circ} X_B$

$X_A = \frac{1}{4}$

$X_B = \frac{3}{4}$

$500 = 200 \times \frac{1}{4} + P_B^{\circ} \times \frac{3}{4}$

$P_B^{\circ} \times \frac{3}{4} = 450$

$P_B^{\circ} = \frac{450 \times 4}{3} = 600$ mm Hg

22. 100 g $CaCO_3$ when reacted with 0.19 mole of HCl then the moles of $CaCl_2$ formed is P × 10^{-3} mol. Find P?

Answer (95)

Sol. $\underset{100\text{ g}}{CaCO_3} + \underset{0.19\text{ mol}}{2HCl} \rightarrow CaCl_2 + H_2CO_3$

= 1 mol

$\therefore$ 1 mol of $CaCO_3$ requires 2 mol HCl

$\therefore$ HCl is limiting request.



$\therefore$ 2 mol HCl $\rightarrow$ 1 mol $CaCl_2$

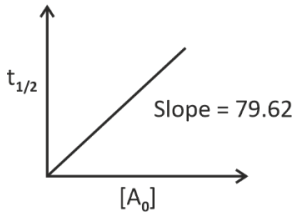
0.19 mol HCl $\rightarrow \frac{1}{2} \times 0.19$ mol $CaCl_2$

= 0.095 mol

= 95 × 10^{-3} mol

P = 95

23. In the following graph between $t_{1/2}$ and initial concentration $[A_0]$. If slope of the graph is 79.62 M^{-1} min. and initial concentration is 2.5 M. Find the concentration of A after 10 min.



Answer (2)

Sol. As $t_{1/2}$ is directly proportional to initial concentration, the reaction follows zero order kinetics.

$$t_{1/2} = \frac{[A_0]}{2k}$$

$$\text{Slope} = \frac{1}{2k}$$

$$\frac{1}{2k} = 79.62$$

$$k = \frac{1}{2 \times 79.62} = 6.28 \times 10^{-3}$$

$$k = \frac{[A_0] - [A_t]}{t}$$

t = 10 min.

$6.28 \times 10^{-3} \times 10 = 2.5 - [A_t]$

$[A_t] = 2.5 - 6.28 \times 10^{-2}$

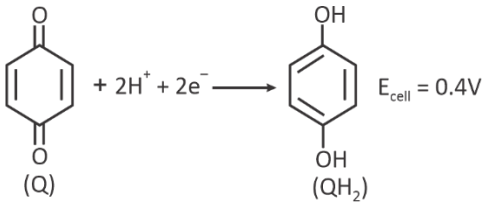
= 2.4372

≈ 2

24. Consider the following electrochemical cell

$Pt|QH_2, Q, H^+(0.1\ M)\ ||\ Ag^+(1\ M)|Ag$

Given that



$E°(Q|QH_2) = 0.7V$; $E°(Ag^+|Ag) = 0.8V$;

$$\frac{2.303\,RT}{F} = 0.06$$

Answer (5)

Sol. Overall cell reaction is

$2Ag^+ + QH_2 \rightarrow 2Ag + Q + 2H^+$

According to Nernst equation

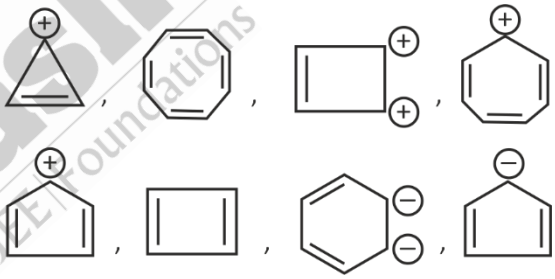
$$E_{cell} = E^\circ_{cell} - \frac{0.06}{2}\log\frac{[H^+]^2}{[Ag^+]^2}$$

$0.4 = 0.1 - 0.06 \log[H^+]$

$$\frac{0.3}{0.06} = pH$$

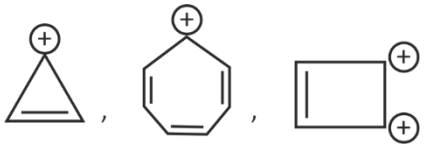
pH = 5

25. Find the total number of aromatic compounds among the following compounds



Answer (4)

Sol. Compound which are cyclic, planar, having cyclic conjugation and follow $(4n + 2)\pi$ electrons rule are aromatic compounds.



aromatic compounds



MATHEMATICS

SECTION - A

Multiple Choice Questions: This section contains 20 multiple choice questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

Choose the correct answer :

1. Let $P_n = \alpha^n + \beta^n$, $P_{10} = 123$, $P_9 = 76$, $P_8 = 47$ and $P_1 = 1$, then quadratic equation whose roots are $\frac{1}{\alpha}$ and $\frac{1}{\beta}$ is

(1) $x^2 + x - 1 = 0$ (2) $x^2 - 2x + 1 = 0$

(3) $x^2 + x - 2 = 0$ (4) $x^2 - x - 2 = 0$

Answer (1)

Sol. $\because P_{10} = P_9 + P_8$

$\Rightarrow P_{10} - P_9 - P_8 = 0$

By Newton's method

Therefore, the equation is

$x^2 - x - 1 = 0$

as $P_1 = 1$

$\Rightarrow \alpha + \beta = 1$ and $\alpha\beta = 1$

$\therefore$ Quadratic equation whose roots are $\frac{1}{\alpha}$ and $\frac{1}{\beta}$ is

$$x^2 - \left(\frac{1}{\alpha} + \frac{1}{\beta}\right)x + \frac{1}{\alpha\beta} = 0$$

$$\Rightarrow x^2 - \left(\frac{\alpha+\beta}{\alpha\beta}\right)x + \frac{1}{\alpha\beta} = 0$$

$\Rightarrow x^2 - (-1)x - 1 = 0$

$\Rightarrow x^2 + x - 1 = 0$

2. Let $a_1, a_2, a_3, \ldots$ is in A.P. and $\sum_{k=1}^{12} a_{2k-1} = -\frac{72}{5}a_1$ and $\sum_{k=1}^{n} a_k = 0$. Then the value of n is

(1) 8 (2) 10

(3) 11 (4) 13

Answer (3)

Sol. $\sum_{k=1}^{12} a_{2k-1} = -\frac{72}{5}a_1$

$$a_1 + a_3 + \cdots + a_{23} = -\frac{72}{5}a_1$$

$$a + a + 2d + \cdots + a + 22d = -\frac{72}{5}a$$

$$12a + 2d(1 + 2 + \cdots 11) = -\frac{72}{5}a$$

$$\Rightarrow 12a + 2d\left(\frac{11\times 12}{2}\right) = -\frac{72}{5}a$$

$$\Rightarrow 132d = -\frac{132}{5}a$$

$\Rightarrow a = -5d$ ...(i)

Also $\sum_{k=1}^{n} a_k = 0$

$\Rightarrow S_n = 0$

$$\Rightarrow \frac{n}{2}[2a + (n-1)d] = 0$$

$\Rightarrow 2a = -(n-1)d$ ...(ii)

From equation (i) and (ii)

$(n-1)d = 10d$

$\therefore n = 11$

3.

4. Given the equation of a hyperbola $H : \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ and its directrix is $x = \sqrt{\frac{10}{81}}$ with a focus at $\left(\sqrt{10}, 0\right)$, then find the value of $9(e + l^2)$, where l is length of latus rectum is

(1) 2697 (2) 2597

(3) 2487 (4) 2587

Answer (4)

THE LEGACY OF SUCCESS CONTINUES
JEE Main (Session-1) 2025
4 STATE TOPPERS | 70+ 100 PERCENTILERS (PHY | CHEM | MATHS) | 1000+ 99 PERCENTILERS & ABOVE | 4000+ 95 PERCENTILERS & ABOVE
100 Percentile | Shreyas Lohiya PSID: 00003389699
100 Percentile | Harsh Jha PSID: 00014863322
99.99 Percentile | Devya Rustagi PSID: 00014768785
99.99 Percentile | Amogh Bansal PSID: 00014769016
OUR JEE CHAMPIONS
Chirag Falor 4 Year Classroom 1 AIR JEE (Adv.) 2020
Tanishka Kabra 4 Year Classroom ALL INDIA FEMALE TOPPER 1 AIR-16 CRL JEE (Adv.) 2022
Sanvi Jain 4 Year Classroom ALL INDIA FEMALE TOPPER 1 AIR-34 CRL JEE (Main) 2024

Sol. $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$

Directrix : $x=\frac{\sqrt{10}}{9}=\frac{a}{e}$ ...(i)

Focus : $(ae,0)\equiv ae=\sqrt{10}$ ...(ii)

(i) × (ii)

$a^2=\frac{10}{9}$

Dividing equation (ii) by (i) we get,

$e^2=9\Rightarrow e=3$

Also, $e^2=9=1+\frac{9b^2}{10}$

$\Rightarrow \quad b^2=\frac{80}{9}$

$l=2\times\frac{b^2}{a}=\frac{2\times\frac{80}{9}}{\frac{\sqrt{10}}{3}}=\frac{160}{3\sqrt{10}}$

Now, $9\left(e+l^2\right)=9\left[3+\frac{25600}{9\times10}\right]$

$9\left(3+\frac{160}{3\sqrt{10}}\right)=27+2560$

= 2587

5. If a twice differentiable function f satisfies $f'(x)=f(x)$ such that $f(0)=\frac{1}{2}=f'(0)$. Then $f''\left(\frac{\pi}{3}\right)$ equals to

(1) $e^{\frac{\pi}{3}}$ (2) $\frac{e^{\frac{\pi}{3}}}{2}$

(3) $\frac{\sqrt{3}}{2}$ (4) $\frac{e^{\frac{2\pi}{3}}}{2}$

Answer (2)

Sol. If $f'(x)=f(x)$

Let $f(x) = y$

$\Rightarrow \quad f'(x)dx=dy$

$\frac{dy}{dx}=y \quad \Rightarrow \ln y=\int dx$

$\Rightarrow \ln y=x+c$

$\Rightarrow y=e^{x+c}=ke^x$, where $k=e^c$

Since $f(0)=\frac{1}{2}$

$\Rightarrow \frac{1}{2}=ke^0 \Rightarrow k=\frac{1}{2}$

$\Rightarrow y=f(x)=\frac{e^x}{2}$

Now, $f'(x)=\frac{e^x}{2}\Rightarrow f'(0)=\frac{1}{2}$ verified

Now, $f''(x)=\frac{e^x}{2}$ at $\frac{\pi}{3}$ will be

$f''\left(\frac{\pi}{3}\right)=\frac{e^{\frac{\pi}{3}}}{2}$

6. Let the system of equations, $3x - y + \beta z = 3$, $2x + \alpha y + z = -3$ and $x + y + 4z = 4$ has infinite solutions, then $22\beta - 9\alpha$ equals to

(1) 165 (2) 164

(3) 163 (4) 162

Answer (2)

Sol. $3x - y + \beta z = 3$

$2x + \alpha y + z = -3$

$x + y = 4z = 4$

has infinite solution

$\Rightarrow \quad \Delta = 0 = \Delta, = \Delta_2 = \Delta_2$

$\Delta=0\Rightarrow\begin{vmatrix}3 & -1 & \beta\\ 2 & \alpha & 1\\ 1 & 1 & 4\end{vmatrix}=0$

$\Rightarrow \quad 2x - xy + 12y + 4 = 0$

$\Delta_2=0\Rightarrow\begin{vmatrix}3 & \beta & 3\\ 2 & 1 & -3\\ 1 & 4 & 4\end{vmatrix}=0$

$\Rightarrow \quad -11\beta + 69 = 0$

$\beta=\frac{69}{11}$



$$\Delta_3 = 0 \Rightarrow \begin{vmatrix} 3 & -1 & 3 \\ 2 & \alpha & -3 \\ 1 & 1 & 4 \end{vmatrix} = 0$$

$9\alpha + 26 = 0$

$$\alpha = -\frac{26}{9}$$

$$22\beta - 9\alpha = 22 \times \frac{69}{11} + \frac{26}{9} \times 9 = 164$$

7. If $\lim\limits_{x \to 0} \frac{(\gamma - 1)e^{x^2} + x^2 \sin(\alpha x)}{\sin(2x) - \beta x} = 3$, then $\alpha + 2\beta + \gamma$ is equal to

(1) 0 (2) 1

(3) 3 (4) 5

Answer (2)

Sol. At $x \to 0$

$\sin 2x - \beta x \to 0$

$\Rightarrow \frac{0}{0}$ form

$\Rightarrow (\gamma - 1)e^0 + 0\sin(\alpha x) \to 0$

$\Rightarrow (\gamma - 1) = 0$

$\Rightarrow \gamma = 1$

$$\Rightarrow \lim_{x \to 0} \frac{x^2 \sin(\alpha x)}{(\sin 2x - \beta x)} = 3$$

$$\Rightarrow \lim_{x \to 0} \frac{x^2\left[\alpha x - \frac{(\alpha x)^3}{3!} + \frac{(\alpha x)^5}{5!} - \cdots\right]}{\left[(2x) - \frac{(2x)^3}{3!} + \frac{(2x)^5}{5!}\right] - \beta x}$$

$$\Rightarrow \lim_{x \to 0} \frac{\alpha x^3 - \frac{\alpha^3 x^5}{3!} + \frac{\alpha^5 x^7}{5!} - \cdots}{x(2 - \beta) - \frac{8x^3}{6} + \frac{2^5 \cdot x^5}{5!} - \cdots} = 3$$

$$\Rightarrow 2 - \beta = 0 \text{ and } \frac{\alpha}{\frac{-8}{6}} = 3$$

$\Rightarrow \beta = 2$

$$\alpha = 3\left(-\frac{8}{6}\right) = -4$$

$\Rightarrow \gamma = 1, \beta = 2, \alpha = -4$

$\Rightarrow \alpha + 2\beta + \gamma = 1 + 4 - 4 = 1$

8. The term independent of x in the binominal expression of $\left(\frac{x+1}{x^{\frac{2}{3}} - x^{\frac{1}{3}} + 1} - \frac{x-1}{x - x^{\frac{1}{2}}}\right)^{10}$ is

(1) 120 (2) 210

(3) 84 (4) 110

Answer (2)

Sol. $(x+1) = \left[\left(x^{\frac{1}{3}}\right) + 1\right]\left[x^{\frac{2}{3}} - x^{\frac{1}{3}} + 1\right]$

$(x+1) = \left(\sqrt{x} - 1\right)\left(\sqrt{x} + 1\right)$

Now

$$\left(\frac{x+1}{x^{\frac{2}{3}} - x^{\frac{1}{3}} + 1}\right) = \left(x^{\frac{1}{3}} + 1\right)$$

and

$$\frac{x-1}{x - x^{\frac{1}{2}}} = \frac{\left(\sqrt{x} - 1\right)\left(\sqrt{x} + 1\right)}{\left(\left(\sqrt{x}\right)^2 - \sqrt{x}\right)}$$

$$= \frac{\left(\sqrt{x} - 1\right)\left(\sqrt{x} + 1\right)}{\left(\sqrt{x}\right)\left(\sqrt{x} - 1\right)} = 1 + \frac{1}{\sqrt{x}}$$

$\Rightarrow$ The expansion become

$$\left[\left(x^{\frac{1}{3}} + 1\right) - \left(1 + \frac{1}{\sqrt{x}}\right)\right]^{10} = \left(x^{\frac{1}{3}} - \frac{1}{x^{\frac{1}{2}}}\right)^{10}$$

The general term of the expansion with be

$$T_{r+1} = {}^{10}C_r \left(-\frac{1}{x^{\frac{1}{2}}}\right)^r \cdot \left(x^{\frac{1}{3}}\right)^{10-r}$$

$$= {}^{10}C_r \frac{(-1)^r}{x^{\frac{r}{2}}} . x^{\left(\frac{10-r}{3}\right)}$$

$$\Rightarrow x^{\left(\frac{10-r}{3} - \frac{r}{2}\right)} . {}^{10}C_r (-1)^r$$

THE LEGACY OF SUCCESS CONTINUES
JEE Main (Session-1) 2025
4 STATE TOPPERS | 70+ 100 PERCENTILERS (PHY | CHEM | MATHS) | 1000+ 99 PERCENTILERS & ABOVE | 4000+ 95 PERCENTILERS & ABOVE
100 Percentile | Shreyas Lohiya | PSID: 00003389699
100 Percentile in Physics | Harsh Jha | PSID: 00014863322
99.99 Percentile | Devya Rustagi | PSID: 00014768785
99.99 Percentile | Amogh Bansal | PSID: 00014769016
OUR JEE CHAMPIONS
Chirag Falor | 4 Year Classroom | 1 AIR JEE (Adv.) 2020
Tanishka Kabra | 4 Year Classroom | ALL INDIA FEMALE TOPPER 1 AIR-16 CRL JEE (Adv.) 2022
Sanvi Jain | 4 Year Classroom | ALL INDIA FEMALE TOPPER 1 AIR-34 CRL JEE (Main) 2024

The term independent when exponent of x is 0

$\Rightarrow \quad \frac{10-r}{3}=\frac{r}{2} \Rightarrow r=4$

$\Rightarrow$ The term independent of x will be

${}^{10}C_4\,(-1)^4x^0 = 210$

9. Let E be an ellipse such that $E:\frac{x^2}{18}+\frac{y^2}{9}=1$. Let point P lies on E such that S and S' are foci of ellipse. Then the sum of min $(PS\cdot PS')$ + max $(PS\cdot PS')$ is

(1) 18 (2) 36

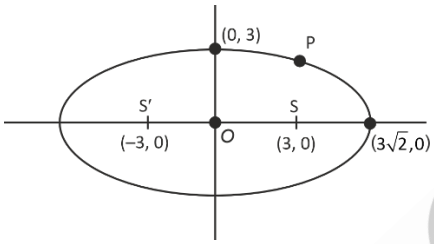
(3) 9 (4) 27

Answer (4)

Sol. For $E=\frac{x^2}{18}+\frac{y^2}{9}=1$.

$a=3\sqrt{2}, b=3 \Rightarrow e=\frac{1}{\sqrt{2}}$

$\Rightarrow$



Since, $PS + PS' = 2a = 6\sqrt{2}$

$\Rightarrow \frac{PS+PS'}{2} \geq \sqrt{PS\cdot PS'}$

$\Rightarrow \left(3\sqrt{2}\right)^2 \geq PS\cdot PS'$

$\Rightarrow \left(PS\cdot PS'\right)_{max} = 18$

and it happens when P lies on the minor axis similarly minima happen P lies on major axis.

$\Rightarrow P \equiv \left(3\sqrt{2}, 0\right)$

$PS = \left(3\sqrt{2}-3\right)$

$PS' = \left(3\sqrt{2}+3\right)$

$\Rightarrow \left(PS\cdot PS'\right)_{min} = \left(3\sqrt{2}\right)^2 - 3^2$

$= 18 - 9 = 9$

$\Rightarrow \left(PS\cdot PS'\right)_{min} + \left(PS\cdot PS'\right)_{max} = 18+9 = 27$

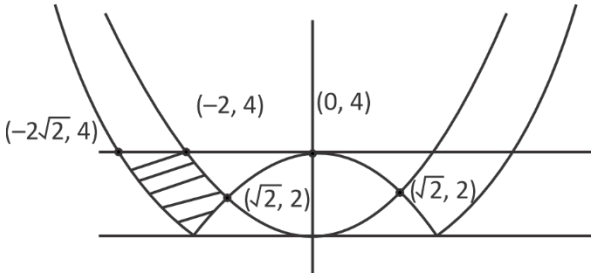
10. The area enclosed by $|4-x^2| \leq y \leq x^2; y \leq 4, x \leq 0$ equals to (in square units)

(1) $\frac{4}{3}(20\sqrt{2}-24)$ (2) $\frac{2}{3}(20\sqrt{2}+24)$

(3) $\frac{4}{3}(10\sqrt{2}+24)$ (4) $\frac{2}{3}(20\sqrt{2}-24)$

Answer (4)

Sol.



Shaded region is the required area.

$$\text{Area} = \int_0^2 \left(-\sqrt{4-y}+\sqrt{4+y}\right)dy + \int_2^4 \left(-\sqrt{y}+\sqrt{4+y}\right)dy$$

$$= \frac{2(4+y)^{3/2}}{3} + \frac{2(4-y)^{3/2}}{3}\Bigg]_0^2 + \frac{2(4+y)^{3/2}}{3} - \frac{2y^{3/2}}{3}\Bigg]_0^4$$

$$= \frac{2}{3}\left[(6^{3/2}+2^{3/2})-(8+8)+(8^{3/2}-8)-(6^{3/2}-2^{3/2})\right]$$

$$= \frac{2}{3}\left[2^{3/2}-16+8^{3/2}-8+2^{3/2}\right]$$

$$= \frac{2}{3}\left[4\sqrt{2}+16\sqrt{2}-24\right]$$

$$= \frac{2}{3}\left[20\sqrt{2}-24\right] \text{ sq. unit}$$

11. Let $\theta \in [-2\pi, 2\pi]$ satisfying $2\cos^2\theta - \sin\theta - 1 = 0$. Then the number of solutions of equation is

(1) 2 (2) 4

(3) 6 (4) 8

Answer (3)

Sol. $2\cos^2\theta - \sin\theta - 1 = 0$

$2(1-\sin^2\theta) - \sin\theta - 1 = 0$

$\Rightarrow \quad -2\sin^2\theta - \sin\theta + 1 = 0$

$\Rightarrow \quad 2\sin^2\theta + \sin\theta - 1 = 0$

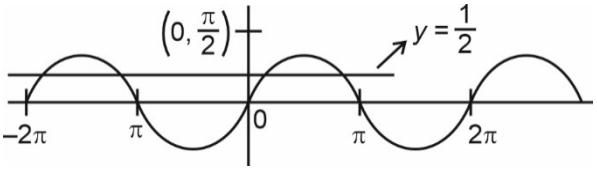
$\Rightarrow \quad 2\sin^2\theta + 2\sin\theta - \sin\theta - 1 = 0$



$\Rightarrow$ $2\sin\theta(\sin\theta+1)-1(\sin\theta+1)=0$

$\Rightarrow$ $(2\sin\theta-1)(\sin\theta+1)=0$

$\Rightarrow$ $\sin\theta=\frac{1}{2},\ -1$



Total 6 solutions possible.

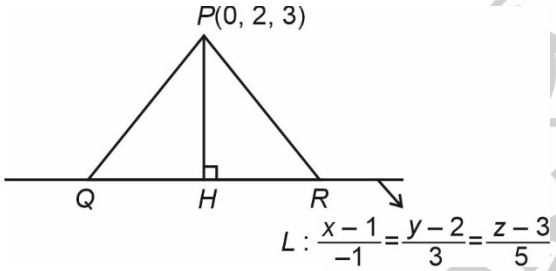
12. If Q and R are two points on line $L:\frac{x-1}{-1}=\frac{y-2}{3}=\frac{z-3}{5}$ such that $QR=5$. If $P(0,2,3)$ be any point, then the area of ΔPQR is

(1) $\sqrt{\frac{85}{14}}$ (2) $\sqrt{\frac{75}{14}}$

(3) $\frac{\sqrt{85}}{14}$ (4) $\frac{\sqrt{75}}{14}$

Answer (1)

Sol. H is point on the line L



$H(-K+1,3K+2,5K+3)$

DR's of PH are $-K+1, 3K, 5K$

$PH\perp L$

$-(-K+1)+3(3K)+5K(5)=0$

$K-1+9K+25K=0$

$35K=1$

$K=\frac{1}{35}$

$H\left(\frac{34}{35},\frac{73}{35},\frac{110}{35}\right)$

$PH=\sqrt{\left(\frac{34}{35}-0\right)^2+\left(\frac{73}{35}-2\right)^2+\left(\frac{110}{35}-3\right)^3}$

$PH=\sqrt{\frac{34}{35}}$

As $(\Delta PQR)=\frac{1}{2}\times PH\times QR$

$=\frac{1}{2}\times\sqrt{\frac{34}{35}}\times 5$

$=\sqrt{\frac{85}{14}}$ sq. unit

13. Let $\sin x\cos y(f(2x+2y)-f(2x-2y))=\cos x\sin y(f(2x+2y)+f(2x-2y))\ \forall\ x,y\in R$ and $f'(0)=\frac{1}{2}$. If $f(x)$ is differentiable function, then $f'''\left(\frac{2\pi}{3}\right)$ is

(1) $\frac{1}{8}$ (2) $\frac{3}{8}$

(3) $-\frac{1}{16}$ (4) $-\frac{3}{4}$

Answer (3)

Sol. $f(2x+2y)[\sin x\cos y-\cos x\sin y]$

$-f(2x-2y)[\sin x\cos y+\cos x\sin y]=0$

$\Rightarrow$ $\sin(x-y)f(2x+2y)=f(2x-2y)\sin(x+y)$

$\Rightarrow$ $\frac{f(2x+2y)}{\sin(x+y)}=\frac{f(2x-2y)}{\sin(x-y)}=k$ (say)

$\therefore$ $f(2x+2y)=k\sin(x+y)$

$f(2x)=k\sin x$ $(\because\ y=0)$

Hence, $f(x)=k\sin\frac{x}{2}$

$f'(x)=\frac{k}{2}\cos\frac{x}{2}$

$f'(0)=\frac{1}{2}\Rightarrow\frac{1}{2}=\frac{k}{2}$

$\Rightarrow$ $k=1$

$\boxed{f(x)=\sin\frac{x}{2}}$

THE LEGACY OF SUCCESS CONTINUES
JEE Main (Session-1) 2025
OUR JEE CHAMPIONS
4 STATE TOPPERS
70+ 100 PERCENTILERS (PHY | CHEM | MATHS)
1000+ 99 PERCENTILERS & ABOVE
4000+ 95 PERCENTILERS & ABOVE
Shreyas Lohiya PSID: 00003389699
Harsh Jha PSID: 00014863322
Devya Rustagi PSID: 00014768785
Amogh Bansal PSID: 00014769016
Chirag Falor 4 Year Classroom 1 AIR JEE (Adv.) 2020
Tanishka Kabra 4 Year Classroom AIR-16 CRL JEE (Adv.) 2022
Sanvi Jain 4 Year Classroom AIR-34 CRL JEE (Main) 2024

$$f'(x)=\frac{1}{2}\cos\frac{x}{2}$$

$$f''(x)=-\frac{1}{4}\sin\frac{x}{2}$$

$$f'''(x)=-\frac{1}{8}\cos\frac{x}{2}$$

$$f'''\left(\frac{2\pi}{3}\right)=-\frac{1}{8}\cos\left(\frac{\pi}{3}\right)$$

$$=-\frac{1}{8}\times\frac{1}{2}=-\frac{1}{16}$$

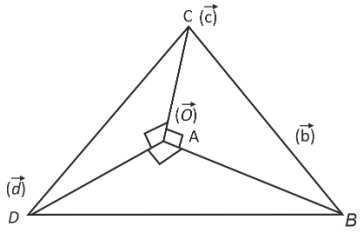
14. For a tetrahedron *ABCD*, the area of triangular face *ABC*, *ACD* and *ABD* is 5, 6 and 7 sq. units respectively. If *AB*, *AC* and *AD* are mutually orthogonal, then the area of triangular face *BCD* is

(1) $\sqrt{11}$ sq. units (2) $\sqrt{110}$ sq. units

(3) $\sqrt{550}$ sq. units (4) $\sqrt{55}$ sq. units.

Answer (2)

Sol.



$\Rightarrow$ Using de gua's theorem

$\Rightarrow [Ar(\Delta BCD)]^2 = Ar(\Delta ABC)^2$

+ Ar(ΔACD)2 + Ar(ΔADB)2

$= 5^2 + 62 + 72 = 110$

$\Rightarrow$ Ar (ΔBCD) = $\sqrt{110}$ sq. units

After:

$$\Delta ABC=\frac{1}{2}\left|\vec{b}\times\vec{c}\right|=\frac{1}{2}\left|\vec{b}\right|\left|\vec{c}\right|$$

$$\Delta ACD=\frac{1}{2}\left|\vec{c}\times\vec{d}\right|=\frac{1}{2}\left|\vec{c}\right|\left|\vec{d}\right|$$

$$\Delta ABD=\frac{1}{2}\left|\vec{b}\times\vec{d}\right|=\frac{1}{2}\left|\vec{b}\right|\left|\vec{d}\right|$$

$$\Rightarrow \text{Area}(\Delta ABD)=\frac{1}{2}\left|(\vec{b}-\vec{d})\times(\vec{c}-\vec{d})\right|$$

$$=\frac{1}{2}\left|\vec{b}\times\vec{c}+\vec{d}\times\vec{b}+\vec{c}\times\vec{d}\right|$$

$$\Rightarrow Ar(BCD)^2=Ar(ABC)^2+Ar(ACD)^2+Ar(ABD)^2$$

15. If $\frac{2+k^2z}{k+k\bar{z}}=z, k\neq 0$, such that $z = x + iy$ and $y \neq 0$ and $|z-1+2i|=1$, then the maximum distance of point $(k + k^2i)$ from the given circle on which z lies is

(1) 2

(2) 4

(3) 3

(4) 1

Answer (2)

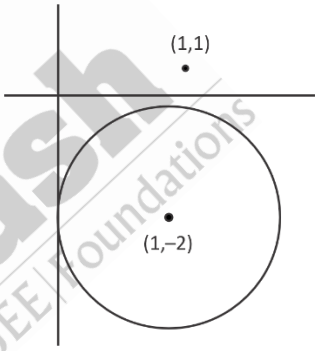
Sol. $2 + k^2z = zk + k|z|^2$

$z(k^2 - k) = k|z|^2 - 2$

If z is not purely real

$\Rightarrow$ $k^2 - k \Rightarrow k = 0$ or 1, $\Rightarrow$ k = 1

$\Rightarrow$ Point is $k + k^2i = (1 + i)$



The maximum distance is

$\sqrt{(1-1)^2+)1+2)^2}$ + radius = 3 + 1 = 4

16. Let C_1 and C_2 are circle passing through (–9, 4), both are in contact with $x + y = 3$ and $x – y = 3$ (tangent lines). If r_1 and r_2 are radius of C_1 and C_2 respectively, then $\left|r_1^2-r_2^2\right|$ equals to

(1) 400

(2) 768

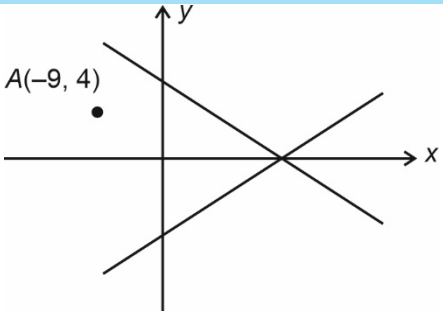
(3) 625

(4) 250

Answer (2)



Sol.



$\because$ $x + y = 3$ and $x - y = 3$ are tangents

$\therefore$ Both circle centre will lie on x-axis

$\therefore$ $(x-a)^2 + y^2 = r^2$

Hence centre is $C(a, 0)$

$r = \sqrt{(a+9)^2 + 16}$ ...(1)

Also $\left|\frac{a-3}{\sqrt{2}}\right| = r$ ...(2)

$\sqrt{(a+9)^2+16} = \left|\frac{a-3}{\sqrt{2}}\right|$

$\Rightarrow$ $a = -5$ or -37

$r = \left|\frac{-5-3}{\sqrt{2}}\right|$ or $\left|\frac{-37-3}{\sqrt{2}}\right|$

$= 4\sqrt{2}$ or $20\sqrt{2}$

$\left|r_1^2 - r_2^2\right| = |32 - 800| = 768$

17. If $(I+A) = \begin{bmatrix} 1 & 0 & a \\ 1 & 1 & 0 \\ a & 2 & 2 \end{bmatrix}$, then the value of det $((a + 1)$ adj $((a-1)A))$ is

(1) $4a^2(a-1)^3(a^2-1)^3$ (2) $8a^3(a^2-1)^6$

(3) $4a^2(a+1)^3(a^2+1)^3$ (4) $4a^2(a^2-1)^3(a+1)^3$

Answer (1)

Sol. $I + A = \begin{bmatrix} 1 & 0 & a \\ 1 & 1 & 0 \\ a & 2 & 2 \end{bmatrix}$

$\Rightarrow$ $A = \begin{bmatrix} 0 & 0 & a \\ 1 & 0 & 0 \\ a & 2 & 1 \end{bmatrix} \Rightarrow |A| = 2a$

det $((a + 1)$ adj $((a-1)A))$

$= (a+1)^3$ det (adj $(a-1)A$)

$= (a+1)^3$ det $((a-1)A)^2$

$= (a+1)^3 [(a-1)^3]^2$ det $(A)^2$

$= (a-1)^3 (a^2-1)^3 |A|^2$

$= 4a^2(a-1)^3(a^2-1)^3$

18. Given $A = \{1, 2, \ldots, 40\}$. Three numbers are randomly selected from set A. Then, the probability that the terms form an increasing G.P is

(1) $\frac{1}{494}$ (2) $\frac{1}{247}$

(3) $\frac{1}{447}$ (4) $\frac{1}{397}$

Answer (1)

Sol. Total cases = ${}^{40}C_3 = 9880$

Since three number a, b and c are in G.P.

$\Rightarrow$ $b^2 = ac$

$\Rightarrow$ $b = \sqrt{ac}$

And this is an increasing G.P. therefore

If $a = 1$ $C = 4, 9, 16, 25, 36$

$a = 2$ $C = 8, 18, 32$

$a = 3$ $C = 12, 27$

$a = 4$ $C = 9, 16, 36$

$a = 5$ $C = 20$

$a = 6$ $C = 24$

$a = 7$ $C = 28$

$a = 8$ $C = 18, 32$

$a = 9$ $C = 36$

$a = 10$ $C = 40$

Total favourable cases = 20

Require probability = $\frac{20}{9880} = \frac{1}{494}$

19.

20.

SECTION - B

Numerical Value Type Questions: This section contains 5 Numerical based questions. The answer to each question should be rounded-off to the nearest integer.

21. The maximum value of n such that 50! is divisible by 3^n is

Answer (22)

THE LEGACY OF SUCCESS CONTINUES
JEE Main (Session-1) 2025
4 STATE TOPPERS
70+ 100 PERCENTILERS (PHY | CHEM | MATHS)
1000+ 99 PERCENTILERS & ABOVE
4000+ 95 PERCENTILERS & ABOVE
100 Percentile in Physics & Maths | Shreyas Lohiya PSID: 00003389699
100 Percentile in Physics | Harsh Jha PSID: 00014863322
99.99 Percentile in Physics & Chemistry | Devya Rustagi PSID: 00014768785
99.99 Percentile | Amogh Bansal PSID: 00014769016
OUR JEE CHAMPIONS
Chirag Falor 4 Year Classroom 1 AIR JEE (Adv.) 2020
Tanishka Kabra 4 Year Classroom ALL INDIA FEMALE TOPPER 1 AIR-16 CRL JEE (Adv.) 2022
Sanvi Jain 4 Year Classroom ALL INDIA FEMALE TOPPER 1 AIR-34 CRL JEE (Main) 2024

Sol. $V_3(50!)=\left[\frac{50}{3}\right]+\left[\frac{50}{9}\right]+\left[\frac{50}{27}\right]+\left[\frac{50}{81}\right]+\cdots$

$= 16 + 5 + 1 + 0 + \cdots$

$= 22$

22. The total number of 10 digits sequences formed by only {0, 1, 2} where 1 should be used at least 5 times and 2 should be used exactly three times, is

Answer (2892)

Sol.

Zero	One	Two		
2	5	3	$\frac{10!}{2!5!3!}-\frac{9!}{5!3!1!}$	= 2520 – 504 = 2016
1	6	3	$\frac{10!}{1!6!3!}-\frac{9!}{6!3!}$	= 840 – 84 = 756
0	7	3	$\frac{10!}{9!3!}$	= 120

Total = 2016 + 756 + 120 = 2892

23. Let $f(x) = 2x^3 + 9x^2a + 12a^2x + 1$ has Local minima and local maxima occur at p & q respectively, such that $p^2 = q$. Then the value of $f(3)$ is

Answer (37)

Sol. $f(x) = 2x^3 + 9x^2a + 12a^2x + 1$

$f'(x) = 6x^2 + 18ax + 12a^2$

$f'(x) = 0$

$x^2 + 3ax + 2a^2 = 0$

$(x + 2a)(x + a) = 0$

$\Rightarrow$ $x = -a, -2a$ [$a \neq 0$ as we will not get maxima and minima at $a = 0$]

Case 1 : When $a > 0$

$f''(x)=12x+18a$

$f''(-a) = -12a + 18a = 6a$

$f''(-2a)=-24a+18a=-6a$

Minima at $x = -a$ & maxima at $x = -2a$

$p = -a$ & $q = -2a$

$p^2 = q$

$a^2 = -2a$

$a = 0, -2$

[Not possible]

Case II : When $a < 0$

$f''(-a) = 6a < 0$

$f''(-2a) = -6a > 0$

Maxima at $x = -a$

and Minima at $x = -2a$

$p = -2a$, $q = -a$

$p^2 = q$

$4a^2 = -a$

$4a^2 + a = 0$

$\Rightarrow$ $\boxed{a=-\frac{1}{4}}$

$f(x)=2x^3+9x^2\left(\frac{-1}{4}\right)+12\left(\frac{-1}{4}\right)^2x+1$

$f(x)=2x^3-\frac{9x^2}{4}+\frac{3x}{4}+1$

$f(3)=54-\frac{81}{4}+\frac{9}{4}+1$

$\Rightarrow$ $\boxed{f(3)=37}$

24. If $\int_0^{e^3}\left[\frac{1}{e^{x-1}}\right]dx=\alpha-\log_e 2$, where $[\cdot]$ is Greatest Integer function, then α^3 equals to

Answer (8)

Sol. $\int_0^{e^3}\left[\frac{1}{e^{x-1}}\right]dx=\int_0^{e^3}\left[e^{1-x}\right]dx$

when, $x = 0$ then $[e^{1-x}] = [e] > 2$

when $e^{1-x} = 2$, then $x = 1 - \ln 2$

and when $e^{1-x} = 1$, then $x = 1$

Now $I=\int_0^{1-\ln 2}2dx+\int_{1-\ln 2}^{1}1dx+\int_1^{e^3}0dx$

$= 2(1 - \ln 2) + (1 - (1 - \ln 2))$

$= 2 - 2\ln 2 + \ln 2$

$= 2 - \ln 2 = \alpha - \ln 2$

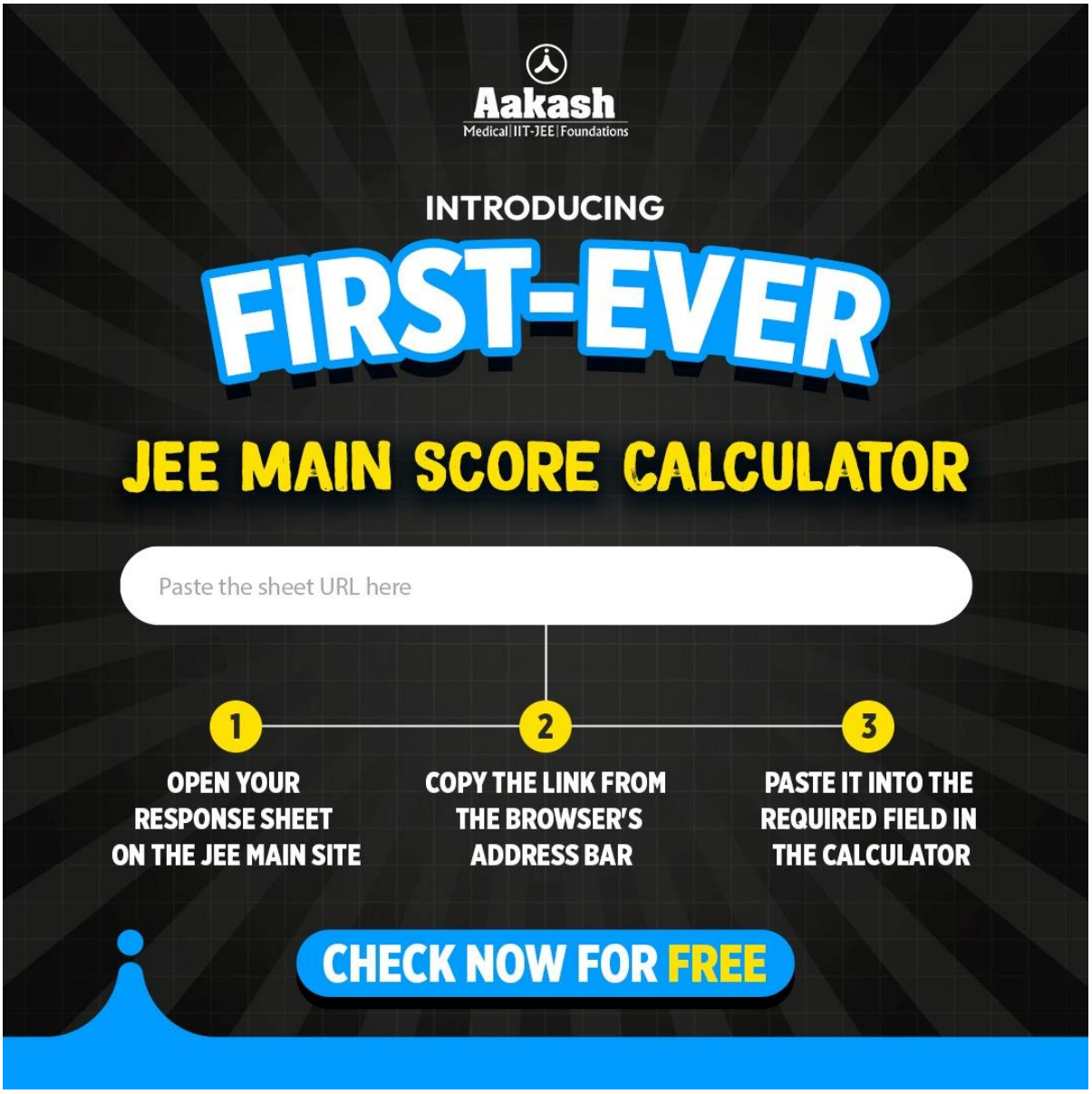
$\Rightarrow \alpha = 2$

$\alpha^3 = 8$

25.

❑ ❑ ❑

THE LEGACY OF SUCCESS CONTINUES
JEE Main (Session-1) 2025
4 STATE TOPPERS | 70+ 100 PERCENTILERS (PHY | CHEM | MATHS) | 1000+ 99 PERCENTILERS & ABOVE | 4000+ 95 PERCENTILERS & ABOVE
100 Percentile in Physics & Maths — Shreyas Lohiya PSID: 00003389699
100 Percentile in Physics — Harsh Jha PSID: 00014863322
99.99 Percentile in Physics & Chemistry — Devya Rustagi PSID: 00014768785
99.99 Percentile — Amogh Bansal PSID: 00014769016
OUR JEE CHAMPIONS
Chirag Falor — 4 Year Classroom — 1 AIR JEE (Adv.) 2020
Tanishka Kabra — 4 Year Classroom — ALL INDIA FEMALE TOPPER — 1 AIR-16 CRL JEE (Adv.) 2022
Sanvi Jain — 4 Year Classroom — ALL INDIA FEMALE TOPPER — 1 AIR-34 CRL JEE (Main) 2024



Curious about your JEE Main score?

Get an instant estimate with our JEE Main Score Calculator -https://bit.ly/4jau6gR